Tutorial-1

31/1/25
Friday

1. Does there exist a simple graph with 7 vertices having degrees 1, 3, 3, 4, 5, 6 & 6?

Ans: Suppose there exist a graph with 7 vertices satisfying the given properties.

Since 2 vertices having degree 6 each of the vertices is adjacent to every other vertices. Hence degree of each vertex is atleast 2. So that the graph has no vertices of degree ~~one~~ 2, which is a contradiction. Hence there does not exist a simple graph with given degree sequence.

2. Is there a simple graph corresponding to the following degree sequence:

a) 1, 1, 2, 3 b) 2, 2, 4, 6

Ans: Here we have 3 vertices of odd degree ie (1, 1, 3)

a) But we know that the no. of odd vertices in a graph must be even. Therefore there does not exist a simple graph corresponding to the graph degree sequence.

b) Here the no. of vertices is 4 & we know that the degree of each vertex cannot exceed 3. (Since the degree of a vertex in a simple graph cannot exceed (n-1)). But here the degree of 2 vertices are 4 & 6.

∴ there does not exist a simple graph having the given degree sequence.

3. Can a simple graph exist with 15 vertices with each of degree 5?

Ans: we know that for any graph, the sum of degrees of vertices is equal to 2x no. of edges. ie which is an even no. Here there are 15 vertices with degree 5. Therefore sum of the degrees is (equal to) $= \Sigma(d_v) = 2e$
$= 15 \times 5 = 75$, which is not an even no.
Since the sum of degrees is not an even no. There does not exist such a graph.

17/1/25
Friday

Tutorial-2

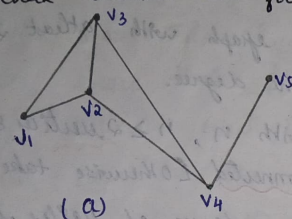
1. Prove that for any simple graph with atleast 2 vertices has, 2 vertices of same degree.
7 marks

Ans: Let G be a simple graph with n, $n \geq 2$ vertices & E edges. Assume that G is connected [Otherwise take any component of G] The Maximum degree of vertex is $n-1$, as G is simple. Being connected, degree cannot be zero. So, the degree of vertex can be any number in the set $\{1, 2, 3 \ldots n-1\}$. Thus, the vertices can have atmost $n-1$ different degrees, but since there are n vertices atleast 2 must have the same degree by pigeon hole principle. If you put more pigeons in fewer pigeon holes, atleast one hole must have more than one pigeon.

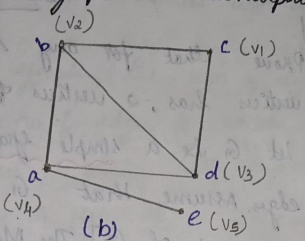
2. Show that in any group of 2 or more people there are 2 always with exactly the same no. of friends inside the group.

Ans: Assume that no 2 people have the same no. of friends in the group. The degree sequence of the vertices will be the set $\{0, 1, 2, 3 \ldots n-1\}$. but a vertex of degree $n-1$ will be connected to each & every other vertex of the graph. This means that no people (vertex) can have no friends, which is a contradiction.

3) Check whether the following graphs are isomorphic.



(a)



(b)

Ans: No. of edges = 6

No. of vertices = 5

$d(V_1) = 2$

$d(V_2) = 3$

$d(V_3) = 3$

$d(V_4) = 3$

$d(V_5) = 1$

No. of edges = 6

No. of vertices = 5

$d(a) = 3$

$d(b) = 3$

$d(c) = 2$

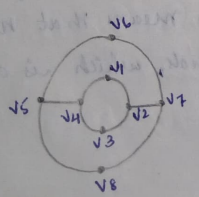
$d(d) = 3$

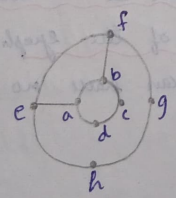
$d(e) = 1$

Clearly there is a one-one correspondance b/w that vertices & b/w their edges such that the incident relationship is preserved.

That is the correspondence b/w the vertices are: V_1, V_2, V_3, V_4, V_5 corresponds to c, b, d, a, e respectively. Therefore the 2 given graphs are isomorphic.

4)





Ans: No. of edges = 10

No. of vertices = 8

$d(V_1) = 2$ $d(V_5) = 3$

$d(V_2) = 3$ $d(V_6) = 2$

$d(V_3) = 2$ $d(V_7) = 3$

$d(V_4) = 3$ $d(V_8) = 2$

No. of edges = 10

No. of vertices = 8

$d(a) = 3$ $d(e) = 3$

$d(b) = 3$ $d(f) = 3$

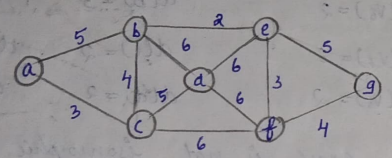
$d(c) = 2$ $d(g) = 2$

$d(d) = 2$ $d(h) = 2$

Here clearly the graph is not isomorphic. In the first graph the vertices with degree 3 form a circuit of length 6 but in second graph circuit is of length 4. & 8.

Tutorial -3

1. State travelling salesman problem, print a travelling salesman tool on the given graph.



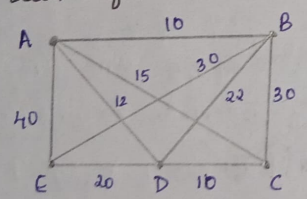
Ans: The T.S.P. is a problem in graph theory requiring the most efficient hamiltonian cycle a salesman can walk through each of n cities.

b — e — g — f — d — c — a — b

$= 2 + 5 + 4 + 6 + 5 + 3 + 5$

$= 30$

4. Find and draw the minimum cost of travelling salesman tour for it. Also mention its cost.



Ans: (i) A — B — E — D — C — A

$10 + 30 + 20 + 10 + 15 = 85$

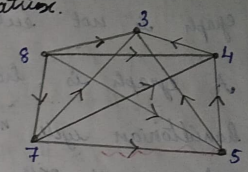
(ii) D — C — A — E — B — D

$10 + 15 + 40 + 30 + 22 = 117$

(i) is the minimum cost of travelling salesman tour for this graph.

5. For a binary relation (is >) on the set $X = \{3, 4, 7, 5, 8\}$; Draw the diagraph representing the above relation. Write its relation matrix.

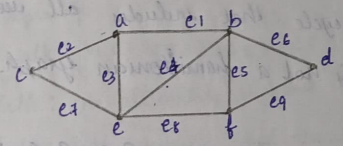
Ans: $x = \{3, 4, 7, 5, 8\}$



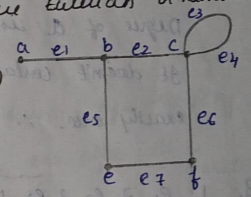
	3	4	7	5	8
3	0	0	0	0	0
4	1	0	0	0	0
5	1	1	0	0	0
7	1	1	1	0	0
8	1	1	1	1	0

2. Check whether the following graphs are Eulerian or hamitonian

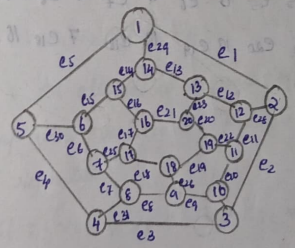
a)



b)



3)



Give the hamiltonian cycle of the following.

2) Ans: An graph is euler if it has an eulerian circuit meaning:

i) All vertices have even degree.

(h) The graph is connected.
The degree of the vertex a & f are odd then the graph is not eulerian.

a) A graph is hamiltonian if there exist a hamiltonian cycle meaning a cycle that visits each vertex exactly once. Possible cycle:

$a\, e_1\, b\, e_6\, d\, e_9\, f\, e_8\, e\, e_7\, c\, e_2\, a$

∴ the graph is hamiltonian.

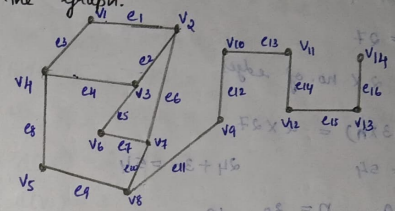
b) Since vertex b & d have odd degrees the graph is not eulerian. Degree of b is ~~one~~ 3.
Degree of d is one.
It doesn't contain a cycle that includes all vertices exactly once. ∴ It is not a hamiltonian graph.

3. 1 e_1 2 e_2 3 e_3 4 e_4 5 e_5 6 e_6 7 e_7 8 e_8 9 e_9 10 e_{10}
11 e_{11} 12 e_{12} 13 e_{23} 20 e_{20} 19 e_{19} 18 e_{18} 17 e_{18} 16 e_{16} 15 e_{14}
14 e_{24} 1

5/2/24
Wednesday

Tutorial - Revision

1. Define walk, path, trail, circuit & fix each of them from the graph.



Ans :→ Walk : A walk is a alternating sequence of vertices & edges, beginning & ending with vertices such that each edge is connected with the vertex preceeding & following it. In a walk, edges & vertices can be repeated.

walk : $v_1\, e_1\, v_2\, e_2\, v_3\, e_5\, v_6\, e_7\, v_7\, e_6\, v_2$

→ Path : An alternating sequence of vertices & edges starting & ending with vertices but no vertices and edges will be repeated.

Path : $v_1\, e_1\, v_2\, e_2\, v_3\, e_4\, v_4$

→ Trail : An alternating sequence of vertices & edges starting & ending with vertices but no edges can be repeated, vertex may be repeated.

Trail : $v_5\, e_9\, v_8\, e_{11}\, v_9\, e_{12}\, v_{10}$.

→ Circuit : A closed path is called a circuit.

circuit : $v_1\, e_1\, v_2\, e_2\, v_3\, e_4\, v_4\, e_3\, v_1$

2. What is the no. of vertices in a undirected connected graph with 27 edges, 6 vertices of degree 4 & 3 vertices of degree 4 & remaining of degree 3?

Ans: No. of edges = 27

Sum of degree = 2 × no. of edges.

$(6\times 2)+(3\times 4)+(3\times n) = 2\times 27$

$12+12+3n = 54$ $\qquad$ $24+3n = 54$

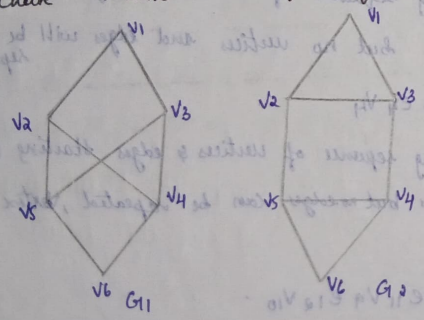
$3n = 30$ $\qquad$ $n = \frac{30}{3} = 10$

∴ Total no. of vertices = 10 + 6 + 3 = 19 vertices

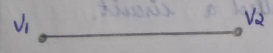
3. Draw a graph which is complete, ~~bip~~ as well as complete bipartite graph.

4. Draw a graph which is eulerian but not hamiltonian.

5. Check whether the following graph are isomorphic or not.

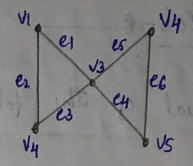


3.



This graph is both complete as well as complete bipartite.

4.



Euler line → $V_1 e_1 V_2 e_2 V_3 e_4 V_4 e_5 V_5 e_6 V_3 e_3 V_1$

5.

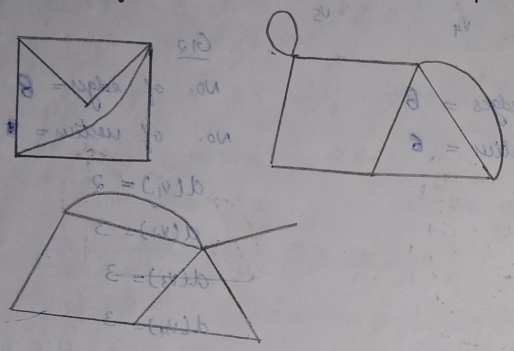
G_1	G_2
No. of edges = 8	No. of edges = 8
No. of vertices = 6	No. of vertices = 6
$d(V_1) = 2$	$d(V_1) = 2$
$d(V_2) = 3$	$d(V_2) = 3$
$d(V_3) = 3$	$d(V_3) = 3$
$d(V_4) = 3$	$d(V_4) = 3$
$d(V_5) = 3$	$d(V_5) = 3$
$d(V_6) = 2$	$d(V_6) = 2$

Here the no. of vertices & no. of edges in G_1 & G_2 are same. Also both does not follows the same degree sequence. Therefore G_1 & G_2 are not isomorphic. ~~Here V_1 in G_1 corresponds to V_1 in G_2, $V_2 \to V_2$, $V_3 \to V_3$, $V_4 \to V_4$, $V_5 \to V_5$, $V_6 \to V_6$.~~

4/3/25
Wednesday

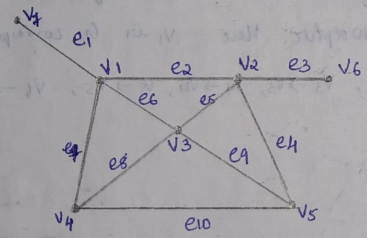
Tutorial - 8

1. Find the geometric dual of the following:



2. Define vertex connectivity of graph & edge connectivity with example.

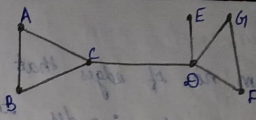
3. Find a spanning tree & hence find all fundamental cut set associated with it, of the the following graph.



5. How many labelled trees with n vertices? Draw all labelled trees with 3 vertices.

6. Define spanning tree & give example.

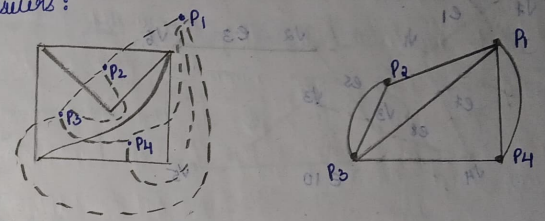
7. Draw a spanning tree for the graph.



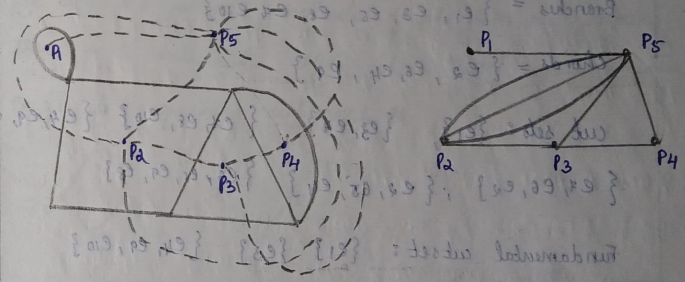
4. Define a planar graph. Is K_4, the complete graph with 4 vertices a planar graph? Justify.

Answers:

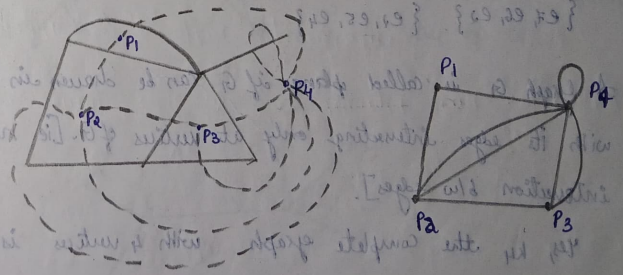
1. a)



b.



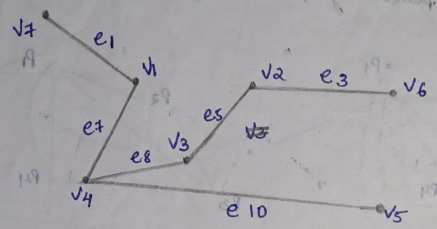
3. c.



2. Edge connectivity :

It refers to the minimum no. of edges that need to be removed from the graph inorder to disconnect the graph.

3.



Branches = $\{e_1, e_3, e_5, e_6, e_7, e_{10}\}$

Chords = $\{e_2, e_8, e_4, e_9\}$

cut sets : $\{e_1\}$, $\{e_3, e_7\}$, $\{e_7, e_8, e_{10}\}$ $\{e_4, e_9, e_{10}\}$

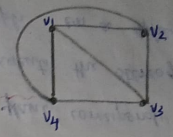
$\{e_7, e_6, e_2\}$; $\{e_2, e_5, e_4\}$ $\{e_6, e_5, e_9, e_8\}$

Fundamental cutset : $\{e_1\}$ $\{e_3\}$ $\{e_4, e_9, e_{10}\}$

$\{e_7, e_6, e_2\}$ $\{e_2, e_5, e_4\}$

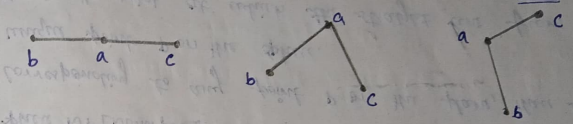
4. A graph G is called planar if G can be drawn in the plane with its edges intersecting only at vertices of G. [ie no intersection b/w edges].

Yes, K_4 the complete graph with 4 vertices is a planar graph. Because it can be drawn in a plane without any edge crossings.



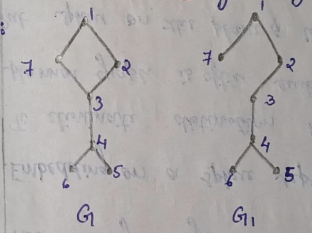
K_4

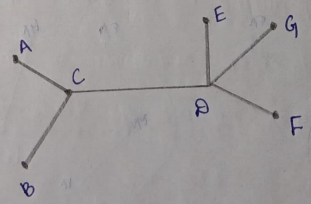
5. The no. of labelled trees with n vertices is n^{n-2}. For $n = 3$, the no. of labelled trees are $3^{3-2} = 3$



6. Let G=(V,E) be an undirected graph, then a spanning tree is a subgraph of G, which itself forms a tree & it is a spanning subgraph.

Eg:



Here G_1 is the spanning tree of G.

7.